Plendl Hourglass Model - FRED edition, run 03 November 2026

Model version 1.0.0. Data: FRED (Federal Reserve Bank of St. Louis), downloaded 2026-11-03. Trend lines fitted over FY2000-FY2026. Projection starts 01 October 2026, monthly, 40 years.

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Headline

• Starting debt D0: $40.525 T (Treasury Debt to the Penny, 2026-10-30); FY2026 year-end debt $40.172 T

• FY2026 receipts: $5.403 T; net interest: $1.360 T; primary deficit: $+0.633 T

• FY2026 interest-to-revenue (I/R): 25.2%; effective rate r0: 3.496%

• Currency in circulation: $2.416 T (2025-10-01), +2.6% year on year

• Observed NIPA interest / receipts (2026-04-01): 21.1% ($1.280 T / $6.056 T SAAR)

Calculated paradigm-shift point

Case	Paradigm-shift point	First month I/R >= 30%	First month I/R >= 40%
Case A: all trend lines fitted 2000-2026	None through September 2066	Sep 2036	Sep 2050
Case B: rate held at FY2026 level (dr = 0)	None through September 2066	Jul 2036	Mar 2050
Case C: rate slope fitted 2021-2026 only (+0.303 pp/yr)	January 2032 (I/R 42.3%)	Apr 2028	Jun 2031

Test: g_q(t) = 12 ln(q_t+1/q_t) and a(t) = g_q(t) - g_q(t-1). The paradigm-shift point is the first month from which, for every remaining month of the horizon, both g_q > 0 and a > 0. 30%/40% are reference lines only.

Fitted trend lines (OLS)

Series	Fit	Parameter	R^2
Receipts	log-linear	g_R = 4.20% / yr	0.917
Gross debt (reference)	log-linear	8.07% / yr	0.991
Effective rate (net interest / avg debt)	linear	dr = -0.0011 pp / yr	0.000
Primary deficit / receipts	linear	b_s = +0.939 pp / yr (start s0 = 11.7%)	0.106

Calculated values, yearly points (Case A; I/R for all cases)

Point	Debt $T	Interest $T	Revenue $T	Primary deficit $T	Rate %	I/R A %	I/R B %	I/R C %
Oct 2026	40.52	1.42	5.40	+0.63	3.50	26.2	26.2	26.2

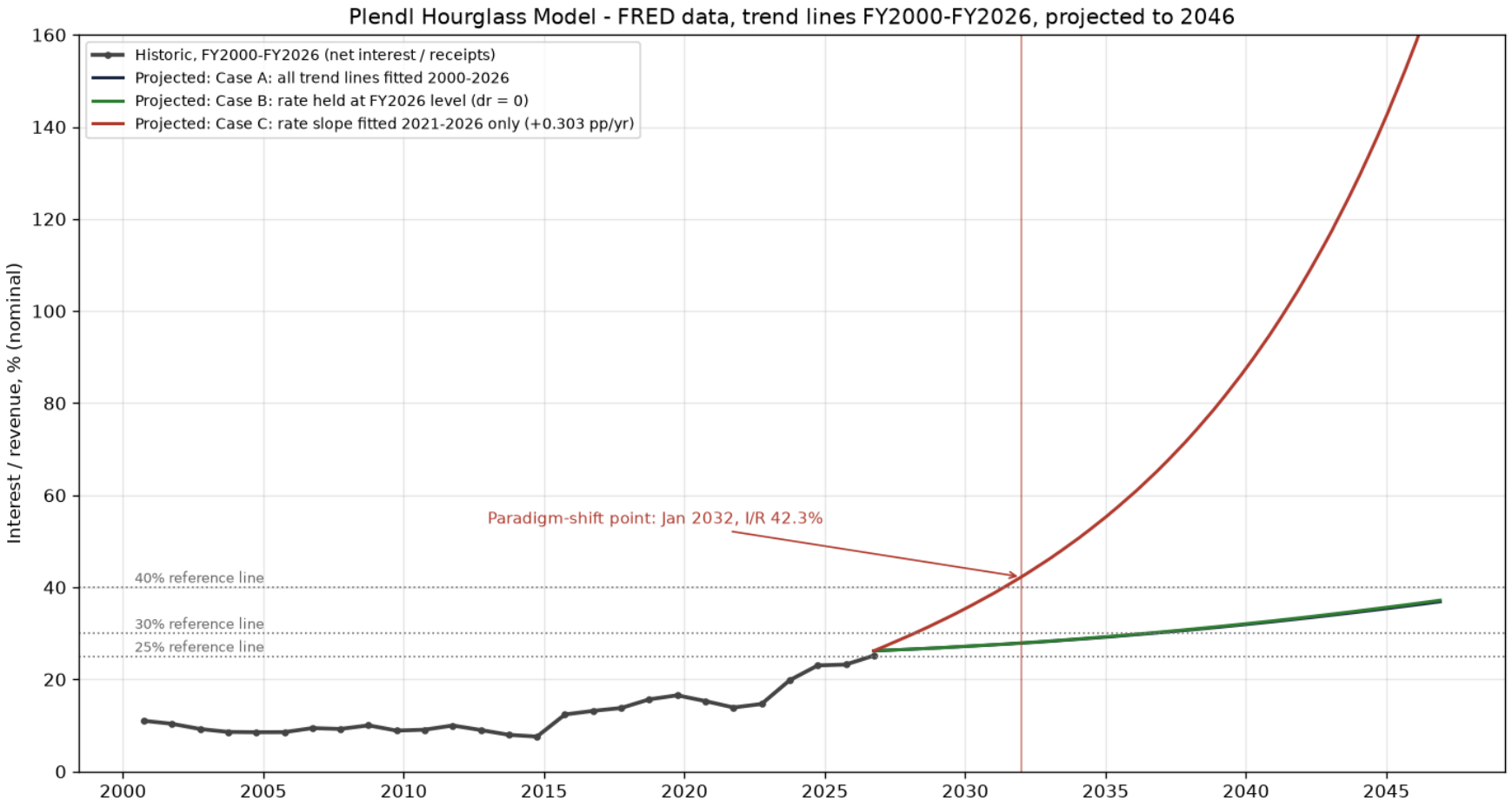
Point	Debt $T	Interest $T	Revenue $T	Primary deficit $T	Rate %	I/R A %	I/R B %	I/R C %
Oct 2027	42.64	1.49	5.63	+0.71	3.49	26.5	26.5	28.8
Oct 2028	44.92	1.57	5.87	+0.80	3.49	26.7	26.8	31.6
Oct 2029	47.37	1.65	6.11	+0.89	3.49	27.1	27.1	34.6
Oct 2030	49.99	1.75	6.37	+0.99	3.49	27.4	27.4	37.8
Oct 2031	52.81	1.84	6.64	+1.09	3.49	27.8	27.8	41.3
Oct 2032	55.84	1.95	6.92	+1.20	3.49	28.2	28.2	45.2
Oct 2033	59.10	2.06	7.21	+1.32	3.49	28.6	28.7	49.4
Oct 2034	62.59	2.18	7.51	+1.44	3.49	29.0	29.1	54.0
Oct 2035	66.33	2.31	7.83	+1.58	3.49	29.5	29.6	59.1
Oct 2036	70.35	2.45	8.16	+1.72	3.48	30.0	30.2	64.7
Oct 2037	74.66	2.60	8.50	+1.87	3.48	30.6	30.7	70.9
Oct 2038	79.28	2.76	8.86	+2.03	3.48	31.2	31.3	77.8
Oct 2039	84.23	2.93	9.23	+2.21	3.48	31.8	31.9	85.5
Oct 2040	89.53	3.12	9.62	+2.39	3.48	32.4	32.6	94.0
Oct 2041	95.22	3.31	10.02	+2.58	3.48	33.0	33.2	103.4
Oct 2042	101.30	3.52	10.44	+2.79	3.48	33.7	33.9	114.0
Oct 2043	107.82	3.75	10.88	+3.01	3.48	34.4	34.7	125.7
Oct 2044	114.79	3.99	11.34	+3.24	3.48	35.2	35.4	138.9
Oct 2045	122.26	4.25	11.82	+3.49	3.47	35.9	36.2	153.6
Oct 2046	130.24	4.52	12.31	+3.75	3.47	36.7	37.0	170.0

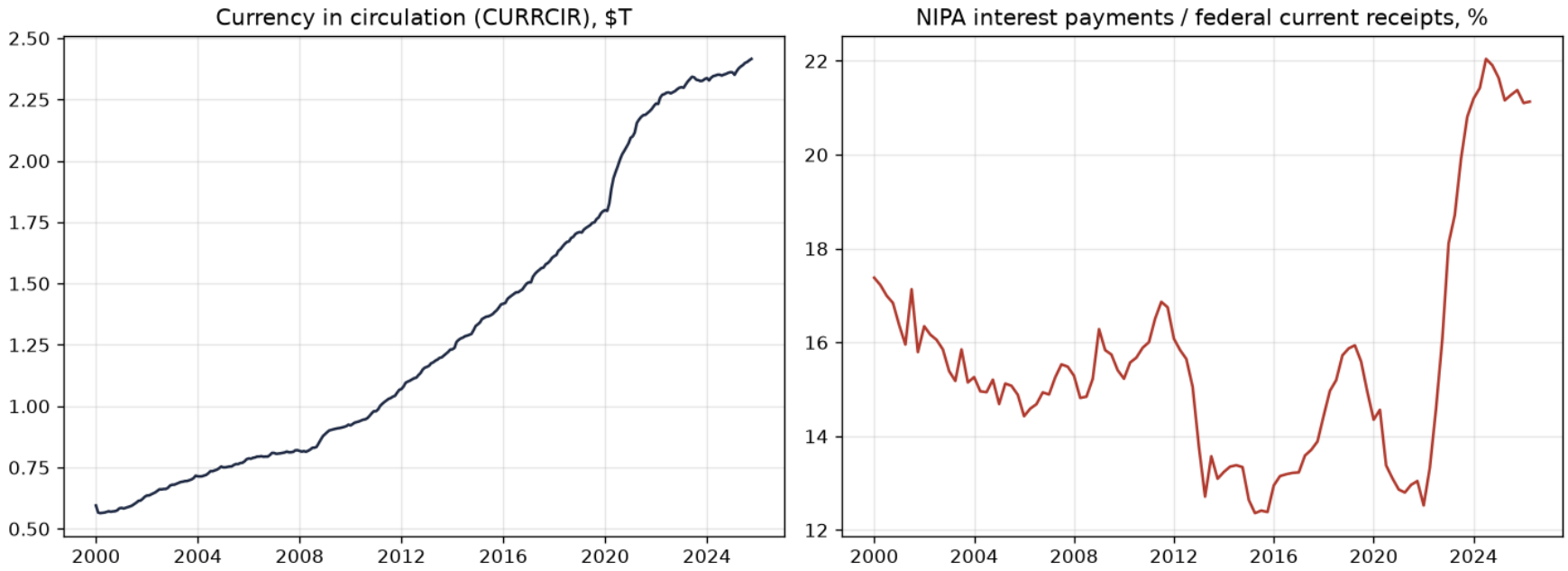
Historic data (FRED)

FY	Receipts $T	Outlays $T	Net interest $T	Primary deficit $T	Gross debt $T	Eff. rate %	I/R %
2000	2.025	1.789	0.223	-0.459	5.67	3.935	11.0
2001	1.991	1.863	0.206	-0.334	5.81	3.591	10.4
2002	1.853	2.011	0.171	-0.013	6.23	2.841	9.2
2003	1.782	2.160	0.153	+0.225	6.78	2.353	8.6
2004	1.880	2.293	0.160	+0.252	7.38	2.263	8.5
2005	2.154	2.472	0.184	+0.134	7.93	2.403	8.5
2006	2.407	2.655	0.227	+0.022	8.51	2.757	9.4
2007	2.568	2.729	0.237	-0.076	9.01	2.708	9.2
2008	2.524	2.983	0.253	+0.206	10.02	2.656	10.0
2009	2.105	3.518	0.187	+1.226	11.91	1.704	8.9
2010	2.163	3.457	0.196	+1.098	13.56	1.541	9.1
2011	2.303	3.603	0.230	+1.070	14.79	1.622	10.0
2012	2.450	3.527	0.220	+0.856	16.07	1.429	9.0

FY	Receipts $T	Outlays $T	Net interest $T	Primary deficit $T	Gross debt $T	Eff. rate %	I/R %
2013	2.775	3.455	0.221	+0.459	16.74	1.347	8.0
2014	3.021	3.506	0.229	+0.256	17.82	1.325	7.6
2015	3.249	3.688	0.402	+0.037	18.15	2.236	12.4
2016	3.267	3.854	0.430	+0.157	19.57	2.280	13.2
2017	3.315	3.981	0.457	+0.209	20.24	2.295	13.8
2018	3.329	4.108	0.522	+0.257	21.52	2.498	15.7
2019	3.462	4.447	0.573	+0.411	22.72	2.590	16.5
2020	3.420	6.552	0.523	+2.609	26.95	2.105	15.3
2021	4.046	6.818	0.562	+2.210	28.43	2.031	13.9
2022	4.896	6.272	0.718	+0.658	30.93	2.418	14.7
2023	4.439	6.134	0.879	+0.816	33.17	2.744	19.8
2024	4.919	6.752	1.133	+0.700	35.46	3.302	23.0
2025	5.235	7.010	1.216	+0.560	37.64	3.326	23.2
2026	5.403	7.396	1.360	+0.633	40.17	3.496	25.2

Note: Debt (all years) = Treasury Debt to the Penny at fiscal-year end; FY2026: $40.172 T (2026-09-30). Starting debt D0 = latest Debt to the Penny, $40.525 T (2026-10-30). For reference, FRED GFDEBTN latest = $39.462 T (2026-04-01, quarter start date) and FYGFD FY2025 = $37.375 T (OMB concept).


Plendl Hourglass Model - FRED data, trend lines FY2000-FY2026, projected to 2046
Historic, FY2000-FY2026 (net interest / receipts)
Projected: Case A: all trend lines fitted 2000-2026
Projected: Case B: rate held at FY2026 level (dr = 0)
Projected: Case C: rate slope fitted 2021-2026 only (+0.303 pp/yr)
Interest / revenue, % (nominal)
Paradigm-shift point: Jan 2032, I/R 42.3%
40% reference line
30% reference line
25% reference line
0
20
40
60
80
100
120
140
160
2000
2005
2010
2015
2020
2025
2030
2035
2040
2045



Data series

- FYFR: Federal Receipts, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYFR
- FYONET: Federal Net Outlays, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYONET
- FYOINT: Federal Outlays: Interest, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYOINT
- FYGFD: Gross Federal Debt, fiscal year end (reference only) (billions $, annual FY) - https://fred.stlouisfed.org/series/FYGFD
- GFDEBTN: Federal Debt: Total Public Debt (reference only) (millions $, quarterly) - https://fred.stlouisfed.org/series/GFDEBTN
- MTSR133FMS: Monthly Treasury Statement: Total Receipts (millions $, monthly) - https://fred.stlouisfed.org/series/MTSR133FMS
- MTSO133FMS: Monthly Treasury Statement: Total Outlays (millions $, monthly) - https://fred.stlouisfed.org/series/MTSO133FMS
- A091RC1Q027SBEA: Federal government current expenditures: Interest payments (billions $ SAAR, quarterly) - https://fred.stlouisfed.org/series/A091RC1Q027SBEA
- FGRECPT: Federal government current receipts (billions $ SAAR, quarterly) - https://fred.stlouisfed.org/series/FGRECPT
- CURRCIR: Currency in Circulation (billions $, monthly) - https://fred.stlouisfed.org/series/CURRCIR
- Treasury Debt to the Penny (total public debt outstanding, daily, $) - https://api.fiscaldata.treasury.gov/services/api/fiscal_service/v2/accounting/od/debt_to_penny